

CUSTOMER BEHAVIOR ANALYSIS REPORT

Shopping Patterns | Revenue Insights | Customer Segmentation

Dataset: 3,900 Customers | 18 Variables | Retail E-Commerce

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1. Executive Summary

This report presents a comprehensive analysis of customer shopping behavior based on a retail dataset comprising 3,900 customers and 18 behavioral attributes. The analysis was conducted using Python (Pandas), SQL queries on PostgreSQL, and Power BI for visualization. Key areas explored include revenue breakdown by demographics, product performance, shipping preferences, discount effectiveness, and customer loyalty segmentation.

3,900 Total Customers Unique customer records	₹233,081 Total Revenue Gross purchase amount	₹59.76 Avg. Spend / Customer Per transaction	3.75 / 5 Avg. Review Rating Product satisfaction	79.9% Loyal Customers >10 previous purchases
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Key Finding: The customer base is overwhelmingly loyal (80%), yet only 27% hold active subscriptions - representing a significant untapped retention and upsell opportunity.

2. Dataset Overview & Methodology

The dataset was sourced from customer transaction records and pre-processed using Python in a Jupyter Notebook environment. The following data preparation steps were applied:

- Missing values in Review Rating were imputed using category-level mean values (37 records affected)
- Column names standardized to snake_case for SQL and Python compatibility
- An age_group categorical variable was derived using quantile-based binning (Young Adult, Adult, Middle-Aged, Senior)
- Purchase frequency was numerically encoded (Weekly = 7 days, Monthly = 30 days, etc.)
- Promo Code Used was found to be a perfect duplicate of Discount Applied and was dropped to eliminate redundancy
- Cleaned data was exported to a PostgreSQL database for SQL-based querying

Dataset Variables

Variable	Type	Description
Customer ID	Integer	Unique customer identifier
Age	Integer (18–70)	Customer age
Gender	Categorical	Male / Female
Item Purchased	Categorical	25 unique product types
Category	Categorical	Clothing, Accessories, Footwear, Outerwear
Purchase Amount (USD)	Integer (₹20–₹100)	Transaction value
Season	Categorical	Fall, Spring, Summer, Winter
Review Rating	Float (2.5–5.0)	Customer product rating
Subscription Status	Binary	Yes / No
Shipping Type	Categorical	Standard, Express, Free, etc.
Discount Applied	Binary	Yes / No
Previous Purchases	Integer (1–50)	Number of prior orders
Payment Method	Categorical	PayPal, Credit Card, Cash, Venmo, etc.
Frequency of Purchases	Categorical	Weekly to Annually

3. Revenue Analysis

3.1 Revenue by Gender

Male customers account for significantly higher purchase revenue compared to female customers, despite both segments being part of the same retail platform.

Gender	Total Revenue (USD)	% of Total
Male	₹157,890	67.7%
Female	₹75,191	32.3%
Total	₹233,081	100%

Male customers generate 2.1x more revenue than female customers. This may reflect a higher proportion of male buyers in the dataset (skewed sampling) or stronger average basket sizes. Further demographic breakdown is recommended.

3.2 Revenue by Age Group

Older customer segments contribute the most revenue, with the 56+ group being the top earner. This challenges the common assumption that younger demographics drive e-commerce sales.

Age Group	Total Revenue (USD)	Rank
56+	₹65,256	1st
46–55	₹45,619	2nd
26–35	₹44,342	3rd
36–45	₹43,234	4th
18–25	₹34,630	5th

3.3 Revenue by Season

Revenue is distributed relatively evenly across seasons, with Fall slightly outperforming. This suggests year-round demand with modest seasonal peaks during autumn months.

Season	Revenue (USD)	% Share
Fall	₹60,018	25.7%
Spring	₹58,679	25.2%
Winter	₹58,607	25.1%
Summer	₹55,777	23.9%

4. Product & Category Analysis

4.1 Revenue by Category

Clothing dominates revenue generation, contributing nearly 45% of total sales. Outerwear is the weakest performer, likely due to its seasonal and niche nature.

Category	Revenue (USD)	Share
Clothing	₹104,264	44.7%
Accessories	₹74,200	31.8%
Footwear	₹36,093	15.5%
Outerwear	₹18,524	7.9%

4.2 Top-Selling Items per Category (by Volume)

Category	Rank 1	Rank 2	Rank 3
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Accessories	Jewelry (171)	Belt (161)	Sunglasses (161)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	—

4.3 Top 5 Products by Review Rating

Item	Avg. Rating	Rank
Gloves	3.86	1st
Sandals	3.84	2nd
Boots	3.82	3rd
Hat	3.80	4th
Skirt	3.79	5th

Accessories and Footwear items (Gloves, Sandals, Boots) dominate the top ratings, suggesting strong product quality and customer satisfaction in these sub-categories relative to higher-revenue Clothing items.

4.4 Items with Highest Discount Rates

The following items had the highest proportion of discount-applied transactions:

Item	Discount Rate (%)	Insight
Hat	50.0%	Highest
Sneakers	49.7%	High
Coat	49.1%	High
Sweater	48.2%	High
Pants	47.4%	High

5. Customer Segmentation

5.1 Loyalty Segmentation

Customers were classified into three segments based on their Previous Purchases count, following the SQL-defined logic:

- NEW: 1 previous purchase
- RETURNING: 2–10 previous purchases
- LOYAL: More than 10 previous purchases

Segment	Count	% of Base
LOYAL	3,116	79.9%
RETURNING	701	18.0%
NEW	83	2.1%

An exceptionally high loyalty rate of ~80% indicates strong retention — yet new customer acquisition is very low at just 2.1%. Marketing investment in top-of-funnel strategies may be warranted to sustain long-term growth.

5.2 Subscription vs. Non-Subscription

Subscription holders represent only 27% of the customer base but are statistically comparable in average spend to non-subscribers, suggesting subscription benefits are not yet driving higher purchase frequency or value.

Subscription	Customers	Total Revenue	Avg. Spend
No	2,847 (73%)	₹170,436	₹59.87
Yes	1,053 (27%)	₹62,645	₹59.49

5.3 Loyal Customers by Subscription Status

Even among the most loyal customers (>10 purchases), the majority remain non-subscribers. This is a critical gap - loyal customers who haven't subscribed represent the easiest conversion opportunity for subscription growth.

Subscription	Loyal Customers	% of Loyal
No	2,247	72.1%
Yes	869	27.9%

6. Shipping & Discount Analysis

6.1 Shipping Type Comparison

Express shipping correlates with marginally higher average purchase amounts compared to Standard shipping, though the difference is not dramatic. Customers choosing faster shipping may also be purchasing higher-value items.

Shipping Type	Avg. Purchase (USD)	Difference
Express	₹60.48	+₹2.02 vs Standard
Standard	₹58.46	Baseline

6.2 Discount Impact on Purchase Amount

Interestingly, customers who received discounts spent slightly less on average than those who did not. This counter-intuitive finding may suggest discounts are applied to lower-ticket items, or that discounted buyers are more price-sensitive.

Discount Applied	Avg. Purchase (USD)	Count
No	₹60.13	2,223
Yes	₹59.28	1,677

Discounts do not appear to drive meaningfully higher basket sizes. A re-evaluation of the discount strategy is recommended - targeting loyal, high-frequency customers with exclusive incentives rather than broad promotional pricing.

7. Payment Method Analysis

Payment preferences are broadly distributed across five methods, indicating a diverse customer base with no single dominant payment channel. This underscores the importance of supporting multiple payment options.

Payment Method	Customers	% Share
PayPal	677	17.4%
Credit Card	671	17.2%
Cash	670	17.2%
Debit Card	636	16.3%
Venmo	634	16.3%

Note: The remaining ~15.5% of customers use methods such as Bank Transfer or other digital wallets not captured in the top 5.

8. Purchase Frequency Distribution

Purchase frequency is nearly uniformly distributed across all seven categories - from Weekly to Annually. This indicates a diverse range of buying habits with no dominant frequency pattern, meaning personalized re-engagement strategies by segment would be more effective than blanket campaigns.

Purchase Frequency	Customer Count	% Share
Every 3 Months	584	15.0%
Annually	572	14.7%
Quarterly	563	14.4%
Monthly	553	14.2%
Bi-Weekly	547	14.0%
Fortnightly	542	13.9%
Weekly	539	13.8%

9. Key Findings & Recommendations

9.1 Summary of Key Findings

- Revenue is male-skewed (68% vs 32%), warranting investigation into gender representation across the customer base.
- Older customers (56+) drive the most revenue despite e-commerce being typically associated with younger shoppers.
- The customer base is 80% loyal, yet only 27% hold subscriptions - the largest untapped retention lever.
- Clothing is the dominant category (45% revenue), with Blouse, Pants, and Shirt as top sellers.
- Accessories items score the highest customer ratings, indicating strong satisfaction in a mid-revenue category.
- Discount application does not meaningfully increase spend - discounts may be misaligned with value drivers.
- Express shipping correlates with slightly higher basket size, suggesting premium customers prefer speed.
- Purchase frequency is evenly spread; no single buying pattern dominates the base.

9.2 Strategic Recommendations

- Convert Loyal Non-Subscribers: Launch a targeted in-app or email campaign for the 2,247 loyal customers without subscriptions, offering a trial or exclusive loyalty-tier benefits.
- Invest in New Customer Acquisition: With only 83 new customers (~2%), increase marketing investment in discovery channels such as social media, search ads, and referral programs.
- Revise Discount Strategy: Shift from broad promotional discounts to personalized, loyalty-driven incentives. Discounts currently do not increase basket size - targeted value-adds (free shipping, loyalty points) may perform better.
- Capitalize on Senior Segment (56+): This group generates the highest revenue. A dedicated UX/content strategy for older shoppers - simpler navigation, trust signals, customer support - could further amplify spending.
- Grow Accessories Category: Despite high satisfaction ratings, Accessories is underrepresented in revenue. Bundling accessories with Clothing purchases or featuring them in seasonal campaigns could boost revenue share.
- Monitor Seasonal Performance: Revenue is nearly flat across seasons. Introducing season-specific limited-edition lines or promotions could generate spikes during lower-performing periods like Summer.

10. Conclusion

This analysis reveals a mature, loyalty-driven customer base with substantial untapped potential in subscription conversion, new customer acquisition, and category expansion. The data signals healthy retention metrics but highlights structural gaps in monetizing loyalty and addressing

demographic imbalances. Integrating these findings into marketing, merchandising, and product strategy will be critical for sustainable revenue growth.

The combination of Python-based EDA, SQL aggregate queries, and Power BI dashboard visualization provides a multi-layered view of customer behavior - enabling both strategic decisions and operational optimizations across the retail value chain.

— End of Report —